

Succeeding in BIO 340 General Genetics

A practical guide to preparing, studying, getting help, and building connections

Prepared by Dr. Jessie Ebie

Genetics builds from week to week. The strongest approach is steady engagement, active practice, and asking for help before small gaps become larger ones.

What to expect

BIO 340 is a four-credit course completed in 7.5 weeks. The average student should plan to spend 24 hours each week learning new material, completing assignments, practicing problems, and reviewing earlier concepts. That is a substantial commitment, so it is important to make room for it in your schedule.

BIO 340 is an upper-level survey of a very large field. The course is challenging but doable if approached correctly. Genetics requires more than memorizing facts or vocabulary. You will interpret information, decide which ideas apply, explain patterns, and work through unfamiliar problems. Many questions are more like word problems or puzzles than recall questions and finding an approach may take more than one attempt.

I am sharing this so you can plan realistically, not to discourage you. Protect time across several days each week, begin early enough to ask questions, and expect problem solving to take time. The course includes some flexibility for when life happens. Outside those options, deadlines are firm. See the course syllabus on Canvas for details.

The learning space

Clinical psychologist Dr. Becky Kennedy uses the term Learning Space™ for the gap between not knowing something and knowing it. This middle can feel frustrating and messy because you are still working things out. That discomfort does not mean you are failing or that you are not good at genetics. It often means you are learning.

In the Learning Space, name what you understand, what remains unclear, and one next step you can attempt. Keep track of your questions and use support hours when you need help moving forward.

Familiarity is not the same as mastery

Material can feel familiar when you reread notes or recognize a worked example. Mastery means you can retrieve the idea without looking, decide when it applies, use it in a new situation, and explain your reasoning.

- Close your notes and solve a problem from the beginning.
- Explain why each step is appropriate, not only what you did.
- Change part of the problem and predict how the answer will change.
- Teach the idea in your own words, then use gaps and mistakes to choose what to practice next.

Study in ways that build mastery

Practice recalling before reviewing

Begin with your notes closed. Write what you remember, draw the process, or try a problem. Then use your notes and course materials to correct and strengthen your understanding. Rereading is more useful after you have identified what you cannot yet recall.

Explain your reasoning step by step

A correct answer is not enough if you cannot explain how you reached it. Write or say why each step makes sense. If you cannot justify a step, you have found a useful question to bring to support hours.

Apply concepts in new situations

Genetics problems may test familiar concepts in unfamiliar contexts. After solving a problem, ask what concept it tested and how the problem could be changed. Try changing the organism, inheritance pattern, given information, or question being asked.

Space and mix your practice

Return to topics over several days and mix related problem types. For example, practice probability, pedigrees, and gene interactions in the same study session. This helps you learn how to select an approach instead of relying on a familiar pattern. This also helps you identify the connections between topics.

Use vocabulary in context

Precise terminology matters because small differences in wording can change a genetics problem. Learn terms by connecting them to diagrams, examples, and relationships rather than memorizing isolated definitions.

Create and explain visual models

Draw and label chromosomes, pathways, crosses, pedigrees, or molecular processes. Flashcards are most helpful when they prompt explanation, comparison, or application rather than recognition alone.

Use your resources

Students who use support resources consistently are often better able to identify gaps early and work through challenging problems. Please do not wait until you feel completely stuck.

- Attend student support hours with Dr. Ebie, a TA, or an ATA.
- Ask questions in your Gene Pool (small learning group) or the Community Forum when appropriate.
- Use ASU's free online tutoring service.
- Use course readings, lectures, recitations, Mastering Genetics practice, and ungraded Dynamic Study Modules intentionally.

Use the Modules in Canvas. Do not rely only on the Canvas To Do list. The To Do list may not include ungraded supporting materials, so working through each module helps make sure you do not miss resources you need for learning and practice.

For additional guidance on metacognition, motivation, and study habits, ASU provides access to [Teach Yourself How to Learn](#) through the ASU Library. This is an excellent resource.

Make support hours work for you

Support hours are a place to learn, not a test of what you already know. You do not need to arrive with a polished question. You may attend any posted session hosted by the instructional team. Sessions are not recorded. If the times do not work, email to ask about an alternative meeting.

If you feel nervous about attending

- Start with one question, one problem attempt, or one place where your reasoning stopped.
- Say, "I am not sure where I went wrong" or "I do not know how to begin." Those are useful starting points.
- Email ahead to share what you want help with or to request a one-on-one conversation. Confusion is expected, and questions and mistakes are part of learning.

Use support hours to build connections

Regular conversations help the instructional team learn how you think and what interests you. Ask about course ideas, future coursework, or careers. Strong engagement may also help you learn about future undergraduate TA opportunities. These roles are not guaranteed, but support hours help you build professional relationships.

Work with classmates productively

Explaining ideas to another student can strengthen recall and reveal gaps. Use your Gene Pool (small learning group) to learn with others, follow each assignment's collaboration rules, and submit your own work.

- Compare reasoning after each person has attempted the problem.
- Ask each other to explain why an approach works.
- Create new examples or questions and keep a shared list of concepts the group wants to ask about.

Know the course policies

Know the policies for deadlines, late passes, exams, collaboration, communication, and academic integrity. The course includes specific flexibility because life happens. Outside those options, deadlines are firm. Check Canvas and your ASU email regularly, see the course syllabus on Canvas for details, and ask if you are unsure how a policy applies.

Prepare for exams through regular practice

- Use recitations and Mastering Genetics problems as practice for applying course concepts.
- Practice without notes so you can see what you can retrieve independently.
- Review mistakes and write what caused each one, such as a vocabulary gap, an incorrect assumption, or a calculation error.
- Practice selecting an approach before calculating or drawing a cross, and begin cumulative review early because later topics rely on earlier ideas.

If you start to fall behind

- Pause and list what is due, what is incomplete, and which concepts are blocking you.
- Prioritize current required work while scheduling a small amount of time to repair the most important earlier gap.
- Review the course syllabus on Canvas so you understand available options, and contact the instructional team early if you need help deciding what to do next.

Disclaimer: Course policies and resources may change. The current syllabus, Canvas course, and assignment directions are the official sources for requirements.